Gihan Shavinda Rathnayake

Cybersecurity Intern | Aspiring Purple Team Analyst | Security Engineering Enthusiast

+94 70 322 2314 | shavindagihan389@gmail.com | Kurunegala, Sri Lanka

LinkedIn | GitHub | Portfolio | Medium | TryHackMe

Summary

B.Sc. (Hons) IT student at the University of Moratuwa (graduation pending) seeking a SOC / Purple Team internship. Built five hands-on security projects covering detection engineering, attack simulation, and detection validation across network, AWS identity, and SOC workflows. Brings secure-API development experience (JWT, RBAC, API security testing) from a professional internship.

Technical Skills

Detection & Purple Team: Detection Engineering, Attack Simulation, Detection Validation, Sigma Rules, MITRE ATT&CK, Threat Hunting, Event Correlation, Incident Response, Sysmon

Network Security: Wireshark, TShark, Tcpdump, Nmap, Suricata, IDS/IPS, TCP/IP

Vulnerability Management: Vulnerability Assessment, CVE/CVSS, EPSS, CISA KEV, Risk Prioritization, Remediation Workflows

Cloud & Identity: AWS IAM, CloudTrail Analysis, Privilege-Escalation Analysis, Neo4j, Google Cloud Platform

Application Security: REST API Security, JWT Authentication, RBAC, Audit Logging, Secure Development

Languages & Tools: Python, JavaScript, PHP, SQL, FastAPI, Laravel, React, Angular, PostgreSQL, MySQL, Docker, Git, GitHub CI/CD, Postman, Linux, Windows, Kali Linux, VMware

Experience

Full-Stack Developer (Intern) Mar 2025 – Nov 2025

Willsj Technologies LLC · Bur Dubai, UAE · Remote

- Implemented JWT authentication and role-based access control (RBAC) across 4 production REST APIs (PHP Laravel / MySQL), restricting access to authorized users and roles.
- Tested APIs for injection and authentication-bypass issues using Postman and API Dog, applying secure-development practices in a Git/GitHub CI/CD workflow.
- Deployed and monitored services on Google Cloud, configuring cloud access controls and application-layer security.

Projects

PurpleLoop – Purple Team Detection, Investigation & SOAR Platform GitHub

- Built a purple team platform (Python, FastAPI, Angular, PostgreSQL) integrating Suricata, Sysmon, Sigma-style detections, and anomaly detection in an isolated VMware lab.
- Correlated attack events into incidents, validated detections against MITRE ATT&CK techniques, and added human-approved SOAR response simulations with RBAC.

VulnForge AI – Continuous Exposure & Remediation Platform GitHub

- Built a vulnerability management platform (FastAPI, React, PostgreSQL, Neo4j, Nmap) that prioritizes risk using CVSS, EPSS, CISA KEV, and attack-path analysis.
- Implemented remediation workflows, reassessment, and regression detection, validated in a private Kali / Metasploitable2 / Windows lab without automatic exploitation.

SOCForge – Purple Team Network Threat Detection Lab GitHub

- Validated 2 detection pipelines (custom Python detector and Suricata IDS) against authorized Nmap reconnaissance in an isolated Kali / Metasploitable2 lab.
- Captured evidence with Tcpdump and TShark, correlated alerts into incidents, and mapped activity to MITRE ATT&CK (T1046) with human-approved response.

CloudHunt – Agentless AWS Threat Detection & Response Platform GitHub

- Analyzed AWS CloudTrail telemetry with behavioral and signature detections, reconstructing AssumeRole chains and ranking incidents by blast radius.
- Modeled IAM privilege-escalation paths in Neo4j and tested detection accuracy on simulated attack scenarios with reversible, human-approved responses.

SOCMind AI – Explainable Security Operations Platform GitHub

- Designed a human-in-the-loop SOC platform with Sigma-style rules, anomaly detection, UEBA, and MITRE ATT&CK mapping that correlates alerts into attack stories.
- Secured it with RBAC/MFA, risk-aware response approvals, and full audit logging of response actions.

Education

University of Moratuwa – B.Sc. (Hons) in Information Technology 2022 – 2026
Faculty of Information Technology | Graduation pending
G.C.E. Advanced Level (2020), Physical Science: Combined Mathematics A, Chemistry A, Physics C

Certifications

- **Google Cybersecurity Certificate** (Coursera) – Foundations of Cybersecurity; Play It Safe: Manage Security Risks; Connect and Protect: Networks and Network Security; Tools of the Trade: Linux and SQL
- **Cisco** – Security Operations Center (SOC)
- **TryHackMe** – Pre Security and Cyber Security 101 Learning Paths

Publications

"Artista – Digital Art Marketplace." ICITR 2024, IEEE (co-author). ieeexplore.ieee.org/document/10857770

References

Dr. K.A. Dilini T. Kulawansa
Senior Lecturer, Department of Computational Mathematics
Faculty of Information Technology, University of Moratuwa
Phone: +94 71 443 4529
Email: dilinik@uom.lk

Randima Lahiru
Associate Tech Lead, 4Axis Technologies Pte Ltd
Phone: +94 71 662 0871
Email: randima@4axissolutions.com